LAB 10 – TRIGGERS

```
DROP TRIGGER IF EXISTS before_emp_insert;
DROP TRIGGER IF EXISTS after_emp_insert;
DROP TRIGGER IF EXISTS before_emp_update;
DROP TRIGGER IF EXISTS after_emp_update;
DROP TRIGGER IF EXISTS before_emp_delete;
DROP TRIGGER IF EXISTS after_emp_delete;

CREATE TABLE
Emp (
emp_id INT
PRIMARY KEY,
emp_name
VARCHAR(30),
dept
VARCHAR(30),
salary INT );

CREATE TABLE Emp_log (     log_id
INT AUTO_INCREMENT PRIMARY KEY,
action VARCHAR(30),     emp_id INT,
log_time TIMESTAMP DEFAULT
CURRENT_TIMESTAMP );

INSERT INTO Emp VALUES
(201, 'Likith Kumar', 'Sales', 52000),
(202, 'Maanas J', 'HR', 43000),
(203, 'Hemanth Gowda', 'IT', 62000),
(204, 'Madhan Kumar', 'IT', 57000),
(205, 'Aditya C', 'Sales', 47000),
(206, 'Dhavan Ganesh', 'HR', 45000);
```

Triggers

```
DELIMITER $$

CREATE TRIGGER before_emp_insert
BEFORE INSERT ON Emp
FOR EACH ROW
BEGIN
    IF NEW.salary <= 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary must be greater than 0';
    END IF;
END$$

CREATE TRIGGER after_emp_insert
AFTER INSERT ON Emp
FOR EACH ROW
BEGIN
    INSERT INTO Emp_log (action, emp_id)
    VALUES ('INSERT', NEW.emp_id);
END$$

CREATE TRIGGER before_emp_update
BEFORE UPDATE ON Emp
FOR EACH ROW
BEGIN
    IF NEW.salary < OLD.salary THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Salary cannot be decreased';
    END IF;
END$$

CREATE TRIGGER after_emp_update
AFTER UPDATE ON Emp
FOR EACH ROW
BEGIN
    INSERT INTO Emp_log (action, emp_id)
    VALUES ('UPDATE', NEW.emp_id);
END$$

CREATE TRIGGER before_emp_delete
```

```
BEFORE DELETE ON Emp
FOR EACH ROW
BEGIN
    IF OLD.salary > 65000 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Cannot delete employee with salary above 65000';
    END IF;
END$$

CREATE TRIGGER after_emp_delete
AFTER DELETE ON Emp
FOR EACH ROW
BEGIN
    INSERT INTO Emp_log (action, emp_id)
    VALUES ('DELETE', OLD.emp_id);
END$$

DELIMITER ;
```

Execute Trigger Examples

```
INSERT INTO Emp VALUES (207, 'Aditya C', 'HR', 39000);
INSERT INTO Emp VALUES (208, 'Madhan Kumar', 'IT', 48000);
UPDATE Emp SET salary = 46000 WHERE emp_id = 202;
UPDATE Emp SET salary = 55000 WHERE emp_id = 201;
DELETE FROM Emp WHERE emp_id = 206;
DELETE FROM Emp WHERE emp_id = 205;

SELECT * FROM Emp;
SELECT * FROM Emp_log;
```

Emp_log output (illustrative log IDs and timestamps)

log_id	action	emp_id	log_time
1	INSERT	207	Generated by database
2	INSERT	208	Generated by database
3	UPDATE	202	Generated by database
4	UPDATE	201	Generated by database
5	DELETE	206	Generated by database
6	DELETE	205	Generated by database

Emp table after the statements above

emp_id	emp_name	dept	salary
201	Likith Kumar	Sales	55000
202	Maanas J	HR	46000
203	Hemanth Gowda	IT	62000
204	Madhan Kumar	IT	57000
207	Aditya C	HR	39000
208	Madhan Kumar	IT	48000